

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5305
Subject Title	Low-altitude Aerial Vehicle Mechanics and Control
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To provide students with the key knowledge relevant to flight mechanics, dynamics, and control for representative aerial vehicles. 2. To provide students with the capacity to formulate the flight control system by using modern engineering tools and algorithms. 3. To provide students with knowledge about intelligent planning and control methods to achieve autonomous flight for manned or unmanned aircraft in low-altitude airspace.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand fundamental concepts of aircraft coordinate systems and forces; b. able to analyse the longitudinal and lateral direction flight mechanics; c. understand classic and modern flight control systems; d. understand search-based and sample-based planning methods and trajectory generation methods; e. understand the working principle of typical onboard electronic equipment; and f. be able to apply the learned knowledge and skills in practice.
Subject Synopsis/ Indicative Syllabus	Dynamics models: Coordinate systems; Kinematic model; Dynamic model; Six Degrees of Freedom (6-DOF) Equations of Motion; Propulsion system model; Model linearization method. Controller design: Classic flight control system; Modern flight control system; State space modelling; Stability, controllability and observability; State feedback design and optimal control. Planning for autonomous flight: Global path planning methods including search-based methods and sample-based methods; Local smooth trajectory generation methods.

	Autopilot system integration and flight simulation: Open-source flight controller; Onboard computer; Robot operating system; Flight simulation platform; Programming and hardware interface; Implementation of key approaches onboard.							
Teaching/Learning Methodology	The teaching and learning methods include lectures and a project. The project is aimed at providing students with the integrated knowledge required for low-altitude aerial vehicles. Technical/practical examples and problems will be raised and discussed in class/hands on sessions.							
	Teaching/Learning Methodology	Outcomes						
		a	b	c	d	e	f	
	Lecture	√	√	√	√	√		
Project	√		√		√	√		
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	1. In-class test	20%	√	√	√	√	√	
	2. Project	40%	√		√		√	√
	3. Examination	40%	√	√	√	√	√	
	Total	100 %						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 0.6 × Continuous Assessment + 0.4 × Final Examination The examination is adopted to assess students on the overall understanding and ability to apply the concepts. It is supplemented by continuous assessment including in-class tests and mini-projects. The continuous assessment is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus. Mini project is used to assess the students’ capacities of self-learning and problem-solving and effective communication skill in English so as to fulfil the requirements of being aircraft control engineers.							
Student Study Effort Expected	Class contact:							
	▪ Lecture					24 Hrs.		
	▪ Project					15 Hrs.		

	Other student study effort:	
	Revision and self-study of materials beyond lectures (3 hours a week for 13 weeks)	39 Hrs.
	Self-study of programming skills used for the mini project (3 hours a week for 10 weeks)	30 Hrs.
	Total student study effort	108 Hrs.
Reading List and References	1. Quan, Quan. Introduction to multicopter design and control. Springer, 2017 2. Randal W. Beard, Timothy W. McLain, Small Unmanned Aircraft: Theory and Practice, Princeton University Press, 2012 3. Thomas R. Yechout, et al, Introduction to aircraft flight mechanics, AIAA, 2012. 4. J. Su, H. Huang, H. Zhang, Y. Wang and F. -Y. Wang, "eVTOL Performance Analysis: A Review From Control Perspectives," in IEEE Transactions on Intelligent Vehicles, 2024, doi: 10.1109/TIV.2024.3387405 5. Z. Shen, G. Zhou, H. Huang, C. Huang, Y. Wang and F. -Y. Wang, "Convex Optimization-Based Trajectory Planning for Quadrotors Landing on Aerial Vehicle Carriers," in IEEE Transactions on Intelligent Vehicles, vol. 9, no. 1, pp. 138-150, Jan. 2024, doi: 10.1109/TIV.2023.3327263.	

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